

Name :- Mihir Mungara

Airflow Assignment – 4

Assignment Definition :- Every day We get 2 different data file in between 4:00 AM to 4:35 AM so utilize Blob storage key Sensor which detects file's Occurrence for both different file location (Check file Occurrence Parrallelly), Once both files are detected aggregate file data and do some transformation then load processed file to some staging blob storage and next task will dump data from blob storage to data base.

You can Integrate some messaging platform like **Slack** to get alerts and notifications about pipeline.

Imported Necessary Libraries and Configured Variables :-

```
from airflow import DAG
from airflow.providers.amazon.aws.sensors.s3 import S3KeySensor
from airflow.providers.amazon.aws.hooks.s3 import S3Hook
from airflow.providers.postgres.hooks.postgres import PostgresHook
from airflow.operators.python import PythonOperator
from airflow.operators.empty import EmptyOperator

from datetime import datetime, timedelta
from io import StringIO
import pandas as pd
import requests
import tempfile

# =====
# CONFIG
# =====

RAW_BUCKET = "raw-bucket-mm"
STAGING_BUCKET = "staging-bucket-mm"

AWS_CONN_ID = "aws_default"
POSTGRES_CONN_ID = "postgres_db_conn"

TEAMS_WEBHOOK = "https://defaultf4814d2338354d87a7dc57a19c0468.4a.environment.api.powerplatform.com:443/"
```

Implementation of Failure Callback :-

```
# =====  
# FAILURE CALLBACK  
# =====  
  
def task_failure_alert(context):  
  
    dag_id = context["dag"].dag_id  
    task_id = context["task_instance"].task_id  
    execution_date = context["execution_date"]  
  
    payload = {  
        "text": (  
            f"✖ Airflow Pipeline Failed\n\n"  
            f"DAG: {dag_id}\n"  
            f"Task: {task_id}\n"  
            f"Execution Date: {execution_date}"  
        )  
    }  
  
    requests.post(TEAMS_WEBHOOK, json=payload)
```

Transformation and Aggregation :-

```
# =====
# TRANSFORM + AGGREGATE
# =====

def aggregate_transform(**kwargs):

    execution_date = kwargs["ds_nodash"]

    s3 = S3Hook(aws_conn_id=AWS_CONN_ID)

    # -----
    # Dynamic file paths
    # -----

    sales_key = f"sales/sales_{execution_date}.csv"

    inventory_key = (
        f"inventory/inventory_{execution_date}.csv"
    )

    # -----
    # Read files from S3
    # -----

    sales_data = s3.read_key(
        key=sales_key,
        bucket_name=RAW_BUCKET
    )

    inventory_data = s3.read_key(
        key=inventory_key,
        bucket_name=RAW_BUCKET
    )

    sales_df = pd.read_csv(
        StringIO(sales_data)
    )

    inventory_df = pd.read_csv(
        StringIO(inventory_data)
    )
```

```
# -----
# Transformations
# -----

final_df = sales_df.merge(
    inventory_df,
    on="product_id",
    how="left"
)

final_df["total_amount"] = (
    final_df["quantity"]
    * final_df["unit_price"]
)

final_df["processing_time"] = (
    pd.Timestamp.now()
)

# -----
# Save temp processed file
# -----

temp_file = tempfile.NamedTemporaryFile(
    delete=False,
    suffix=".csv"
)

final_df.to_csv(
    temp_file.name,
    index=False
)

kwargs["ti"].xcom_push(
    key="processed_file_path",
    value=temp_file.name
)
```

Uploading to Staging :-

```
# =====
# UPLOAD TO STAGING S3
# =====

def upload_to_staging(**kwargs):

    execution_date = kwargs["ds_nodash"]

    ti = kwargs["ti"]

    file_path = ti.xcom_pull(
        task_ids="aggregate_transform",
        key="processed_file_path"
    )

    s3 = S3Hook(
        aws_conn_id=AWS_CONN_ID
    )

    staging_key = (
        f"processed/"
        f"processed_sales_"
        f"{execution_date}.csv"
    )

    s3.load_file(
        filename=file_path,
        key=staging_key,
        bucket_name=STAGING_BUCKET,
        replace=True
    )

    ti.xcom_push(
        key="staging_key",
        value=staging_key
    )
```

Load to Postgres :-

```
# =====
# LOAD INTO POSTGRES
# =====

def load_to_postgres(**kwargs):

    ti = kwargs["ti"]

    file_path = ti.xcom_pull(
        task_ids="aggregate_transform",
        key="processed_file_path"
    )

    (variable) postgres_hook: Any
    postgres_hook = PostgresHook(
        postgres_conn_id=POSTGRES_CONN_ID
    )

    conn = postgres_hook.get_conn()
    cursor = conn.cursor()

    cursor.execute("""
        CREATE TABLE IF NOT EXISTS sales_staging (
            order_id INT,
            product_id VARCHAR(50),
            customer_id VARCHAR(50),
            quantity INT,
            unit_price FLOAT,
            order_timestamp TIMESTAMP,
            product_name VARCHAR(255),
            category VARCHAR(100),
            stock_available INT,
            warehouse VARCHAR(100),
            last_updated TIMESTAMP,
            total_amount FLOAT,
            processing_time TIMESTAMP
        );
    """)

    conn.commit()
```

```
with open(file_path, "r") as f:
    cursor.copy_expert(
        """
        COPY sales_staging
        FROM STDIN
        WITH CSV HEADER
        """,
        f
    )

    conn.commit()
    cursor.close()
    conn.close()
```

Notification Configuration :-

```
# =====
# SUCCESS NOTIFICATION
# =====

def notify_success(**kwargs):

    dag_id = kwargs["dag"].dag_id
    run_id = kwargs["run_id"]
    logical_date = kwargs["logical_date"]

    payload = {
        "@type": "MessageCard",
        "@context": "https://schema.org/extensions",
        "themeColor": "00FF00",
        "title": "✅ Airflow Pipeline Success",
        "text": (
            f"""DAG:** {dag_id}\n\n"""
            f"""Run ID:** {run_id}\n\n"""
            f"""Logical Date:** {logical_date}"""
        )
    }

    requests.post(
        TEAMS_WEBHOOK,
        json=payload
    )
```

Dag Config :-

```
# =====
# DAG
# =====

default_args = {
    "owner": "mihir",
    "depends_on_past": False,
    "retries": 2,
    "retry_delay": timedelta(minutes=5),
    "on_failure_callback": task_failure_alert
}

with DAG(
    dag_id="Assignment-4",
    description="S3 sensor ETL pipeline",
    start_date=datetime(2026, 1, 1),
    schedule="0 4 * * *",
    catchup=False,
    default_args=default_args,
    tags=["s3", "etl", "pipeline"]
) as dag:

    start = EmptyOperator(
        task_id="start"
    )
```

Parallel File Detection Code :-

```
# =====  
# PARALLEL FILE DETECTION  
# =====  
  
wait_sales_file = S3KeySensor(  
    task_id="wait_sales_file",  
  
    bucket_name=RAW_BUCKET,  
  
    bucket_key=(  
        "sales/"  
        "sales_{{ ds_nodash }}.csv"  
    ),  
  
    wildcard_match=False,  
  
    aws_conn_id=AWS_CONN_ID,  
  
    poke_interval=60,  
  
    timeout=2100,  
  
    mode="reschedule"  
)  
  
wait_inventory_file = S3KeySensor(  
    task_id="wait_inventory_file",  
  
    bucket_name=RAW_BUCKET,  
  
    bucket_key=(  
        "inventory/"  
        "inventory_{{ ds_nodash }}.csv"  
    ),  
  
    wildcard_match=False,  
  
    aws_conn_id=AWS_CONN_ID,  
  
    poke_interval=60,  
  
    timeout=2100,  
  
    mode="reschedule"  
)
```

ETL Tasks :-

```
# =====  
# ETL TASKS  
# =====  
  
aggregate_task = PythonOperator(  
    task_id="aggregate_transform",  
    python_callable=aggregate_transform  
)  
  
upload_task = PythonOperator(  
    task_id="upload_to_staging",  
    python_callable=upload_to_staging  
)  
  
postgres_task = PythonOperator(  
    task_id="load_to_postgres",  
    python_callable=load_to_postgres  
)  
  
notify_task = PythonOperator(  
    task_id="notify_success",  
    python_callable=notify_success  
)  
  
end = EmptyOperator(  
    task_id="end"  
)
```

Dependencies :-

```
# =====  
# DEPENDENCIES  
# =====  
  
start >> [  
    wait_sales_file,  
    wait_inventory_file  
]  
  
[  
    wait_sales_file,  
    wait_inventory_file  
] >> aggregate_task  
  
(  
    aggregate_task  
    >> upload_task  
    >> postgres_task  
    >> notify_task  
    >> end  
)
```


Data in S3 Bucket :-

--> Bucket Structure

raw-bucket-mm Info

[Objects](#) [Metadata](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [File systems - new](#) [Access Points](#)

Objects (2) [Refresh](#) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

☐	Name	Type	Last modified	Size	Storage class
☐	inventory/	Folder	-	-	-
☐	sales/	Folder	-	-	-

--> Inventory Folder

inventory/ [Copy S3 URI](#)

[Objects](#) [Properties](#)

Objects (1) [Refresh](#) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

☐	Name	Type	Last modified	Size	Storage class
☐	inventory_20260511.csv	csv	May 11, 2026, 10:53:37 (UTC+05:30)	560.0 B	Standard

```
inventory_20260511.csv X
inventory_20260511.csv
1 product_id,product_name,category,stock_available,warehouse,last_updated
2 P101,Gold Ring,Rings,25,Ahmedabad_WH,2026-05-11 04:02:10
3 P102,Silver Necklace,Necklaces,40,Surat_WH,2026-05-11 04:03:45
4 P103,Diamond Earrings,Earrings,18,Mumbai_WH,2026-05-11 04:05:12
```

--> Sales Folder

sales/ [Copy S3 URI](#)

[Objects](#) [Properties](#)

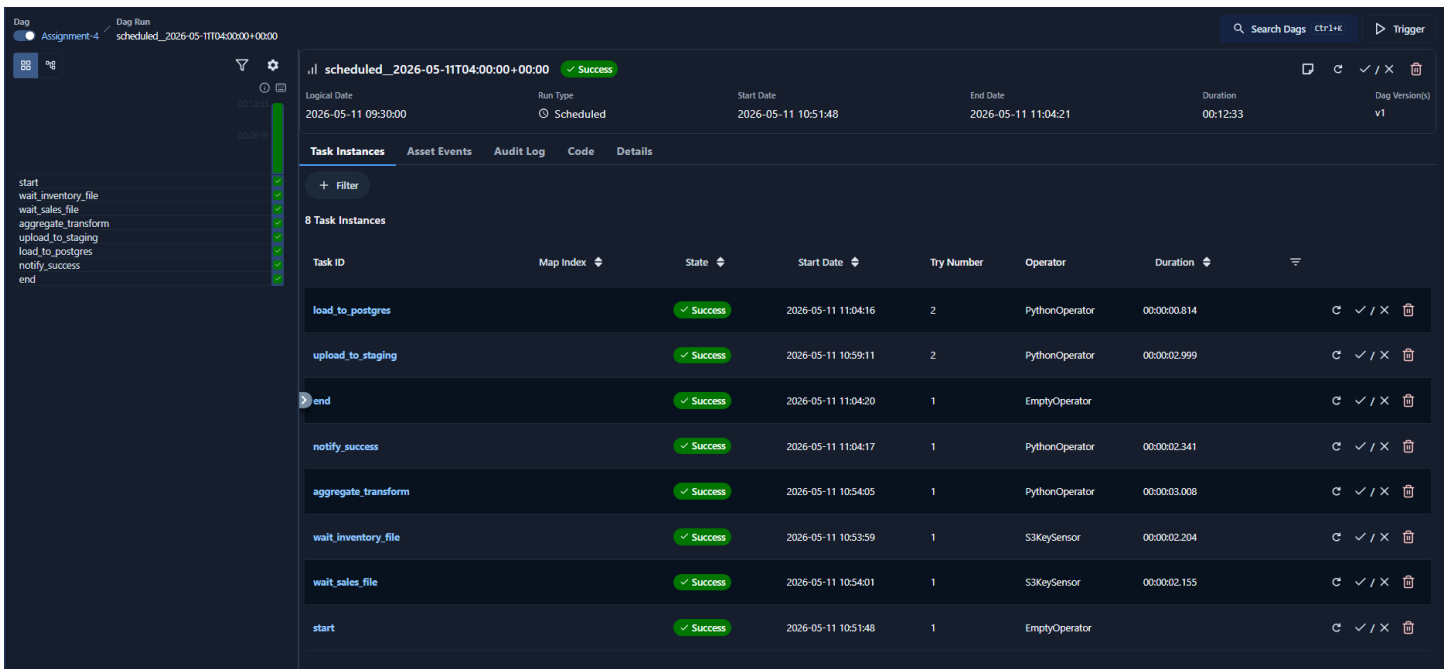
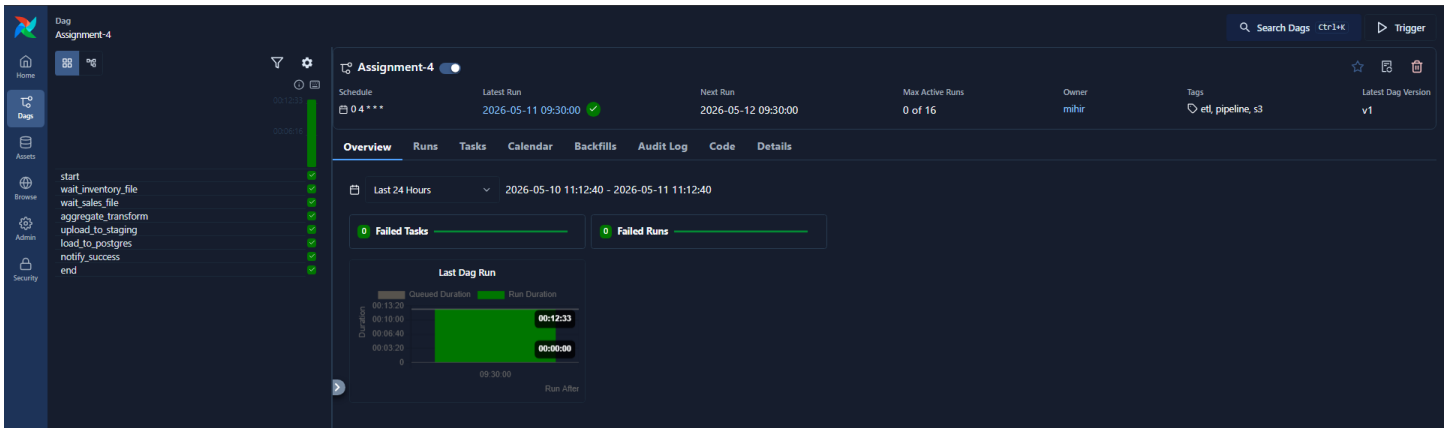
Objects (1) [Refresh](#) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

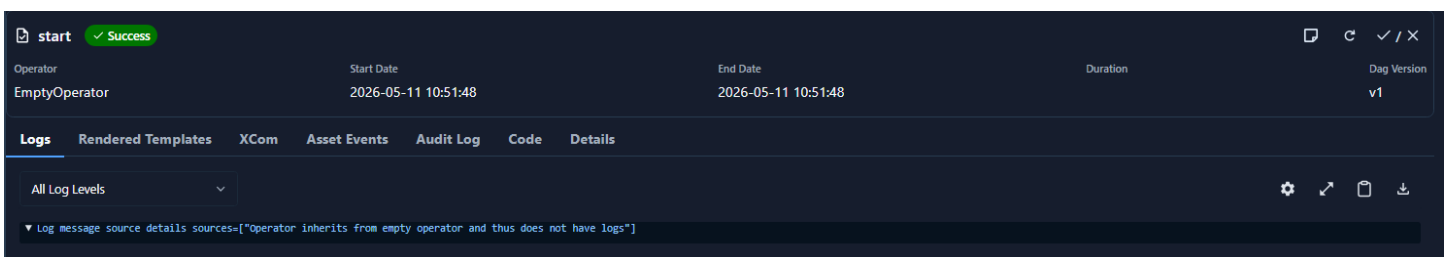
☐	Name	Type	Last modified	Size	Storage class
☐	sales_20260511.csv	csv	May 11, 2026, 10:53:53 (UTC+05:30)	490.0 B	Standard

```
sales_20260511.csv X
C: > Users > mihir.mungara > Desktop > spark > sales_20260511.csv
1 order_id,product_id,customer_id,quantity,unit_price,order_timestamp
2 1001,P101,C501,2,450,2026-05-11 04:05:12
3 1002,P102,C502,1,1200,2026-05-11 04:07:30
4 1003,P103,C503,3,300,2026-05-11 04:10:05
5 1004,P101,C504,1,450,2026-05-11 04:15:18
```

Airflow UI showing Assignment-4 :-



Logs for Start Task :-



Logs For Wait_Inventory_File Task :-

wait_inventory_file

Success

Operator

S3KeySensor

Start Date

2026-05-11 10:53:59

End Date

2026-05-11 10:54:01

Duration

00:00:02.204

Dag Version

v1

Logs

Rendered Templates

XCom

Asset Events

Audit Log

Code

Details

All Log Levels

All Sources

Log message source details sources=["/opt/airflow/logs/dag_id=Assignment-4/run_id=scheduled_2026-05-11T04:00+00:00/task_id=wait_inventory_file/attempt=1.log"]

0

[2026-05-11 10:51:49]

INFO

- DAG bundles loaded: dags-folder

1

[2026-05-11 10:51:49]

INFO

- Filling up the DagBag from /opt/airflow/dags/Assignment4.py

2

[2026-05-11 10:51:52]

WARNING

- The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.python.PythonOperator'. category=UserWarning

3

[2026-05-11 10:51:52]

WARNING

- The 'airflow.operators.empty.EmptyOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.empty.EmptyOperator'. category=UserWarning

4

[2026-05-11 10:51:52]

INFO

- Poking for key : s3://raw-bucket-mm/inventory/inventory_20260511.csv

5

[2026-05-11 10:51:52]

INFO

- AWS Connection (conn_id='aws_default', conn_type='aws') credentials retrieved from login and password.

6

[2026-05-11 10:51:52]

WARNING

- ProvidersManager.hooks is deprecated. Use ProvidersManagerTaskRuntime.hooks from task-sdk instead. category=DeprecationWarning

7

[2026-05-11 10:51:55]

INFO

- Rescheduling task, marking task as UP_FOR_RESCHEDULE

8

[2026-05-11 10:52:56]

INFO

- DAG bundles loaded: dags-folder

9

[2026-05-11 10:52:56]

INFO

- Filling up the DagBag from /opt/airflow/dags/Assignment4.py

10

[2026-05-11 10:52:57]

WARNING

- The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.python.PythonOperator'. category=UserWarning

11

[2026-05-11 10:52:57]

WARNING

- The 'airflow.operators.empty.EmptyOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.empty.EmptyOperator'. category=UserWarning

12

[2026-05-11 10:52:57]

INFO

- Poking for key : s3://raw-bucket-mm/inventory/inventory_20260511.csv

13

[2026-05-11 10:52:57]

INFO

- AWS Connection (conn_id='aws_default', conn_type='aws') credentials retrieved from login and password.

14

[2026-05-11 10:52:57]

WARNING

- ProvidersManager.hooks is deprecated. Use ProvidersManagerTaskRuntime.hooks from task-sdk instead. category=DeprecationWarning

15

[2026-05-11 10:52:58]

INFO

- Rescheduling task, marking task as UP_FOR_RESCHEDULE

16

[2026-05-11 10:53:59]

INFO

- DAG bundles loaded: dags-folder

17

[2026-05-11 10:53:59]

INFO

- Filling up the DagBag from /opt/airflow/dags/Assignment4.py

18

[2026-05-11 10:54:00]

WARNING

- The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.python.PythonOperator'. category=UserWarning

19

[2026-05-11 10:54:00]

WARNING

- The 'airflow.operators.empty.EmptyOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.empty.EmptyOperator'. category=UserWarning

20

[2026-05-11 10:54:00]

INFO

- Poking for key : s3://raw-bucket-mm/inventory/inventory_20260511.csv

21

[2026-05-11 10:54:00]

INFO

- AWS Connection (conn_id='aws_default', conn_type='aws') credentials retrieved from login and password.

22

[2026-05-11 10:54:00]

WARNING

- ProvidersManager.hooks is deprecated. Use ProvidersManagerTaskRuntime.hooks from task-sdk instead. category=DeprecationWarning

23

[2026-05-11 10:54:01]

INFO

- Success criteria met. Exiting.

Logs For Wait_Sales_File Task :-

wait_sales_file

Success

Operator

S3KeySensor

Start Date

2026-05-11 10:54:01

End Date

2026-05-11 10:54:03

Duration

00:00:02.155

Dag Version

v1

Logs

Rendered Templates

XCom

Asset Events

Audit Log

Code

Details

All Log Levels

All Sources

Log message source details sources=["/opt/airflow/logs/dag_id=Assignment-4/run_id=scheduled_2026-05-11T04:00+00:00/task_id=wait_sales_file/attempt=1.log"]

0

[2026-05-11 10:51:49]

INFO

- DAG bundles loaded: dags-folder

1

[2026-05-11 10:51:49]

INFO

- Filling up the DagBag from /opt/airflow/dags/Assignment4.py

2

[2026-05-11 10:51:52]

WARNING

- The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.python.PythonOperator'. category=UserWarning

3

[2026-05-11 10:51:52]

WARNING

- The 'airflow.operators.empty.EmptyOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.empty.EmptyOperator'. category=UserWarning

4

[2026-05-11 10:51:52]

INFO

- Poking for key : s3://raw-bucket-mm/sales/sales_20260511.csv

5

[2026-05-11 10:51:52]

INFO

- AWS connection (conn_id='aws_default', conn_type='aws') credentials retrieved from login and password.

6

[2026-05-11 10:51:52]

WARNING

- ProvidersManager.hooks is deprecated. Use ProvidersManagerTaskRuntime.hooks from task-sdk instead. category=DeprecationWarning

7

[2026-05-11 10:51:58]

INFO

- Rescheduling task, marking task as UP_FOR_RESCHEDULE

8

[2026-05-11 10:52:59]

INFO

- DAG bundles loaded: dags-folder

9

[2026-05-11 10:52:59]

INFO

- Filling up the DagBag from /opt/airflow/dags/Assignment4.py

10

[2026-05-11 10:52:59]

WARNING

- The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.python.PythonOperator'. category=UserWarning

11

[2026-05-11 10:52:59]

WARNING

- The 'airflow.operators.empty.EmptyOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.empty.EmptyOperator'. category=UserWarning

12

[2026-05-11 10:52:59]

INFO

- Poking for key : s3://raw-bucket-mm/sales/sales_20260511.csv

13

[2026-05-11 10:53:00]

INFO

- AWS connection (conn_id='aws_default', conn_type='aws') credentials retrieved from login and password.

14

[2026-05-11 10:53:00]

WARNING

- ProvidersManager.hooks is deprecated. Use ProvidersManagerTaskRuntime.hooks from task-sdk instead. category=DeprecationWarning

15

[2026-05-11 10:53:01]

INFO

- Rescheduling task, marking task as UP_FOR_RESCHEDULE

16

[2026-05-11 10:54:01]

INFO

- DAG bundles loaded: dags-folder

17

[2026-05-11 10:54:01]

INFO

- Filling up the DagBag from /opt/airflow/dags/Assignment4.py

18

[2026-05-11 10:54:02]

WARNING

- The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.python.PythonOperator'. category=UserWarning

19

[2026-05-11 10:54:02]

WARNING

- The 'airflow.operators.empty.EmptyOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.empty.EmptyOperator'. category=UserWarning

20

[2026-05-11 10:54:02]

INFO

- Poking for key : s3://raw-bucket-mm/sales/sales_20260511.csv

21

[2026-05-11 10:54:02]

INFO

- AWS connection (conn_id='aws_default', conn_type='aws') credentials retrieved from login and password.

22

[2026-05-11 10:54:02]

WARNING

- ProvidersManager.hooks is deprecated. Use ProvidersManagerTaskRuntime.hooks from task-sdk instead. category=DeprecationWarning

23

[2026-05-11 10:54:03]

INFO

- Success criteria met. Exiting.

Logs For Aggregate_Transform Task :-

aggregate_transform

Success

Operator

PythonOperator

Start Date

2026-05-11 10:54:05

End Date

2026-05-11 10:54:08

Duration

00:00:03.008

Dag Version

v1

Logs

Rendered Templates

XCom

Asset Events

Audit Log

Code

Details

All Log Levels

All Sources

Log message source details sources=["/opt/airflow/logs/dag_id=Assignment-4/run_id=scheduled__2026-05-11T04:00:00+00:00/task_id=aggregate_transform/attempt=1.log"]

0

[2026-05-11 10:54:05]

INFO

- DAG bundles loaded: dags-folder

1

[2026-05-11 10:54:05]

INFO

- Filling up the DagBag from /opt/airflow/dags/Assignment4.py

2

[2026-05-11 10:54:05]

WARNING

- The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.python.PythonOperator'. category=UserWarning

3

[2026-05-11 10:54:05]

WARNING

- The 'airflow.operators.empty.EmptyOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.empty.EmptyOperator'. category=UserWarning

4

[2026-05-11 10:54:05]

INFO

- AWS Connection (conn_id='aws_default', conn_type='aws') credentials retrieved from login and password.

5

[2026-05-11 10:54:08]

INFO

- Done. Returned value was: None

Logs For Upload_to_Staging Task :-

upload_to_staging

Success

Operator

PythonOperator

Try Number

2

Start Date

2026-05-11 10:59:11

End Date

2026-05-11 10:59:14

Duration

00:00:02.999

Dag Version

v1

Logs

Rendered Templates

XCom

Asset Events

Audit Log

Code

Details

Task Tries

1

✖

2

✔

All Log Levels

All Sources

Log message source details sources=["/opt/airflow/logs/dag_id=Assignment-4/run_id=scheduled__2026-05-11T04:00:00+00:00/task_id=upload_to_staging/attempt=2.log"]

0

[2026-05-11 10:59:11]

INFO

- DAG bundles loaded: dags-folder

1

[2026-05-11 10:59:11]

INFO

- Filling up the DagBag from /opt/airflow/dags/Assignment4.py

2

[2026-05-11 10:59:12]

WARNING

- The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.python.PythonOperator'. category=UserWarning

3

[2026-05-11 10:59:12]

WARNING

- The 'airflow.operators.empty.EmptyOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.empty.EmptyOperator'. category=UserWarning

4

[2026-05-11 10:59:12]

INFO

- AWS Connection (conn_id='aws_default', conn_type='aws') credentials retrieved from login and password.

5

[2026-05-11 10:59:12]

WARNING

- ProvidersManager.hooks is deprecated. Use ProvidersManagerTaskRuntime.hooks from task-sdk instead. category=DeprecatedImportWarning

6

[2026-05-11 10:59:14]

INFO

- Done. Returned value was: None

Logs for Load_to_Postgres Task :-

load_to_postgres

Success

Operator

PythonOperator

Try Number

2

Start Date

2026-05-11 11:04:16

End Date

2026-05-11 11:04:17

Duration

00:00:00.814

Dag Version

v1

Logs

Rendered Templates

XCom

Asset Events

Audit Log

Code

Details

Task Tries

1

✖

2

✔

All Log Levels

All Sources

Log message source details sources=["/opt/airflow/logs/dag_id=Assignment-4/run_id=scheduled__2026-05-11T04:00:00+00:00/task_id=load_to_postgres/attempt=2.log"]

0

[2026-05-11 11:04:16]

INFO

- DAG bundles loaded: dags-folder

1

[2026-05-11 11:04:16]

INFO

- Filling up the DagBag from /opt/airflow/dags/Assignment4.py

2

[2026-05-11 11:04:17]

WARNING

- The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.python.PythonOperator'. category=UserWarning

3

[2026-05-11 11:04:17]

WARNING

- The 'airflow.operators.empty.EmptyOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.empty.EmptyOperator'. category=UserWarning

4

[2026-05-11 11:04:17]

INFO

- Done. Returned value was: None

Logs for Notify_Succes Task :-

notify_success

Success

Operator

PythonOperator

Start Date

2026-05-11 11:04:17

End Date

2026-05-11 11:04:19

Duration

00:00:02.341

Dag Version

v1

Logs

Rendered Templates

XCom

Asset Events

Audit Log

Code

Details

All Log Levels

All Sources

Log message source details sources=["/opt/airflow/logs/dag_id=Assignment-4/run_id=scheduled__2026-05-11T04:00:00+00:00/task_id=notify_success/attempt=1.log"]

0

[2026-05-11 11:04:17]

INFO

- DAG bundles loaded: dags-folder

1

[2026-05-11 11:04:17]

INFO

- Filling up the Dagbag from /opt/airflow/dags/Assignment4.py

2

[2026-05-11 11:04:19]

WARNING

- The 'airflow.operators.python.PythonOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.python.PythonOperator'. category=UserWarning

3

[2026-05-11 11:04:19]

WARNING

- The 'airflow.operators.empty.EmptyOperator' attribute is deprecated. Please use 'airflow.providers.standard.operators.empty.EmptyOperator'. category=UserWarning

4

[2026-05-11 11:04:19]

INFO

- Done. Returned value was: None

Logs For End Task :-

end

Success

Operator

EmptyOperator

Start Date

2026-05-11 11:04:20

End Date

2026-05-11 11:04:20

Duration

Dag Version

v1

Logs

Rendered Templates

XCom

Asset Events

Audit Log

Code

Details

All Log Levels

Log message source details sources=["Operator inherits from empty operator and thus does not have logs"]

Output/Staging Bucket (showing bucket and it’s file content generated by dag run) :-

staging-bucket-mm

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems - new

Access Points

Objects (1)

Copy S3 URI

Copy URL

Download

Open L

Delete

Actions

Create folder

Upload

Find objects by prefix

Name

processed/

Type

Folder

Last modified

-

Size

-

Storage class

-

processed/

Copy S3 URI

Objects

Properties

Objects (1)

Copy S3 URI

Copy URL

Download

Open L

Delete

Actions

Create folder

Upload

Find objects by prefix

Name

processed_sales_20260511.csv

Type

csv

Last modified

May 11, 2026, 10:59:14 (UTC+05:30)

Size

1.4 KB

Storage class

Standard

processed_sales_20260511.csv

X

C: > Users > mihir.mungara > Downloads > processed_sales_20260511.csv

1

order_id,product_id,customer_id,quantity,unit_price,order_timestamp,product_name,category,stock_available,warehouse,last_updated,total_amount,processing_time

2

1001,P101,C501,2,450,2026-05-11 04:05:12,Gold Ring,Rings,25,Ahmedabad_WH,2026-05-11 04:02:10,900,2026-05-11 05:24:08.032956

3

1002,P102,C502,1,1200,2026-05-11 04:07:30,Silver Necklace,Necklaces,40,Surat_WH,2026-05-11 04:03:45,1200,2026-05-11 05:24:08.032956

4

1003,P103,C503,3,300,2026-05-11 04:10:05,Diamond Earrings,Earrings,18,Mumbai_WH,2026-05-11 04:05:12,900,2026-05-11 05:24:08.032956

5

1004,P101,C504,1,450,2026-05-11 04:15:18,Gold Ring,Rings,25,Ahmedabad_WH,2026-05-11 04:02:10,450,2026-05-11 05:24:08.032956

6

1005,P104,C505,2,700,2026-05-11 04:20:44,Bracelet,Bracelets,30,Ahmedabad_WH,2026-05-11 04:06:23,1400,2026-05-11 05:24:08.032956

7

1006,P105,C506,1,1500,2026-05-11 04:22:19,Luxury Watch,Watches,12,Delhi_WH,2026-05-11 04:09:44,1500,2026-05-11 05:24:08.032956

8

1007,P102,C507,4,1200,2026-05-11 04:25:11,Silver Necklace,Necklaces,40,Surat_WH,2026-05-11 04:03:45,4800,2026-05-11 05:24:08.032956

9

1008,P106,C508,2,950,2026-05-11 04:29:53,Pearl Pendant,Necklaces,20,Jaipur_WH,2026-05-11 04:11:30,1900,2026-05-11 05:24:08.032956

10

1009,P103,C509,1,300,2026-05-11 04:31:10,Diamond Earrings,Earrings,18,Mumbai_WH,2026-05-11 04:05:12,300,2026-05-11 05:24:08.032956

11

1010,P107,C510,5,200,2026-05-11 04:34:59,Anklet,Accessories,55,Ahmedabad_WH,2026-05-11 04:12:19,1000,2026-05-11 05:24:08.032956

12

Showing Table via PostgreSQL Docker Container :-

```
PS C:\Users\mihir_mungara\Desktop\Airflow_Practice> docker exec -it airflow_practice-postgres-1 psql -U airflow
psql (13.23 (Debian 13.23-1.pgdg13+1))
Type "help" for help.

airflow=# SELECT * FROM sales_staging;
 order_id | product_id | customer_id | quantity | unit_price | order_timestamp | product_name | category | stock_available | warehouse | last_updated | total_amount | processing_time
-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----
 1801 | P101 | C501 | 2 | 450 | 2026-05-11 04:05:12 | Gold Ring | Rings | 25 | Ahmedabad_WH | 2026-05-11 04:02:10 | 900 | 2026-05-11 05:24:34.436475
 1802 | P102 | C502 | 1 | 1200 | 2026-05-11 04:07:30 | Silver Necklace | Necklaces | 40 | Surat_WH | 2026-05-11 04:03:45 | 1200 | 2026-05-11 05:24:34.436475
 1803 | P103 | C503 | 3 | 300 | 2026-05-11 04:10:05 | Diamond Earrings | Earrings | 18 | Mumbai_WH | 2026-05-11 04:05:12 | 900 | 2026-05-11 05:24:34.436475
 1804 | P101 | C504 | 1 | 450 | 2026-05-11 04:15:18 | Gold Ring | Rings | 25 | Ahmedabad_WH | 2026-05-11 04:02:10 | 450 | 2026-05-11 05:24:34.436475
 1805 | P104 | C505 | 2 | 700 | 2026-05-11 04:20:44 | Bracelet | Bracelets | 30 | Ahmedabad_WH | 2026-05-11 04:06:23 | 1400 | 2026-05-11 05:24:34.436475
 1806 | P105 | C506 | 1 | 1500 | 2026-05-11 04:22:19 | Luxury Watch | Watches | 12 | Delhi_WH | 2026-05-11 04:09:44 | 1500 | 2026-05-11 05:24:34.436475
 1807 | P102 | C507 | 4 | 1200 | 2026-05-11 04:25:11 | Silver Necklace | Necklaces | 40 | Surat_WH | 2026-05-11 04:03:45 | 4800 | 2026-05-11 05:24:34.436475
 1808 | P106 | C508 | 2 | 950 | 2026-05-11 04:29:53 | Pearl Pendant | Necklaces | 20 | Jaipur_WH | 2026-05-11 04:11:30 | 1900 | 2026-05-11 05:24:34.436475
 1809 | P103 | C509 | 1 | 300 | 2026-05-11 04:31:10 | Diamond Earrings | Earrings | 18 | Mumbai_WH | 2026-05-11 04:05:12 | 300 | 2026-05-11 05:24:34.436475
 1810 | P107 | C510 | 5 | 200 | 2026-05-11 04:34:59 | Anklet | Accessories | 55 | Ahmedabad_WH | 2026-05-11 04:12:19 | 1000 | 2026-05-11 05:24:34.436475
 1801 | P101 | C501 | 2 | 450 | 2026-05-11 04:05:12 | Gold Ring | Rings | 25 | Ahmedabad_WH | 2026-05-11 04:02:10 | 900 | 2026-05-11 05:24:08.032956
 1802 | P102 | C502 | 1 | 1200 | 2026-05-11 04:07:30 | Silver Necklace | Necklaces | 40 | Surat_WH | 2026-05-11 04:03:45 | 1200 | 2026-05-11 05:24:08.032956
 1803 | P103 | C503 | 3 | 300 | 2026-05-11 04:10:05 | Diamond Earrings | Earrings | 18 | Mumbai_WH | 2026-05-11 04:05:12 | 900 | 2026-05-11 05:24:08.032956
 1804 | P101 | C504 | 1 | 450 | 2026-05-11 04:15:18 | Gold Ring | Rings | 25 | Ahmedabad_WH | 2026-05-11 04:02:10 | 450 | 2026-05-11 05:24:08.032956
 1805 | P104 | C505 | 2 | 700 | 2026-05-11 04:20:44 | Bracelet | Bracelets | 30 | Ahmedabad_WH | 2026-05-11 04:06:23 | 1400 | 2026-05-11 05:24:08.032956
 1806 | P105 | C506 | 1 | 1500 | 2026-05-11 04:22:19 | Luxury Watch | Watches | 12 | Delhi_WH | 2026-05-11 04:09:44 | 1500 | 2026-05-11 05:24:08.032956
 1807 | P102 | C507 | 4 | 1200 | 2026-05-11 04:25:11 | Silver Necklace | Necklaces | 40 | Surat_WH | 2026-05-11 04:03:45 | 4800 | 2026-05-11 05:24:08.032956
 1808 | P106 | C508 | 2 | 950 | 2026-05-11 04:29:53 | Pearl Pendant | Necklaces | 20 | Jaipur_WH | 2026-05-11 04:11:30 | 1900 | 2026-05-11 05:24:08.032956
 1809 | P103 | C509 | 1 | 300 | 2026-05-11 04:31:10 | Diamond Earrings | Earrings | 18 | Mumbai_WH | 2026-05-11 04:05:12 | 300 | 2026-05-11 05:24:08.032956
 1810 | P107 | C510 | 5 | 200 | 2026-05-11 04:34:59 | Anklet | Accessories | 55 | Ahmedabad_WH | 2026-05-11 04:12:19 | 1000 | 2026-05-11 05:24:08.032956
(20 rows)
```

Notification on Teams under Mentioned Channel :-

